

SAGE UNIVERSITY INDORE



LAB MANUAL

Academic Year: 2024-25

Session: Jan-June 2025

Semester: II

Faculty Name :
Department : Computer Application
Subject code :
Subject Name : Computer Networks Lab

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Experiment- 01

Study of Different Types of LAN & Network Equipment

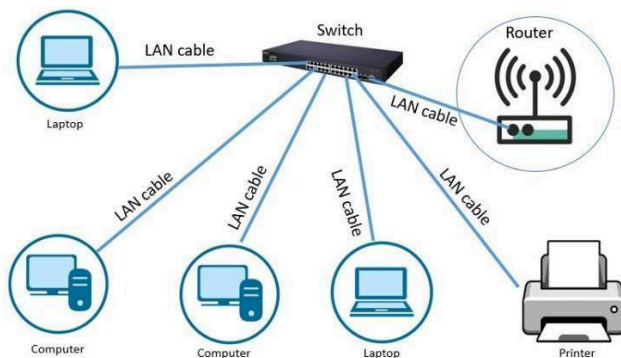
Lab Objective: Understanding various types of LAN (Local Area Networks) and network equipment commonly used in networking environments.

Prerequisites: Basic knowledge of networking concepts and familiarity with common networking devices

Outcome: Gain practical experience in identifying and understanding different types of LAN and network equipment.

Description:

A Local Area Network (LAN) is a network that connects computers and devices in a limited geographical area, such as a home, school, computer laboratory, office building, or closely positioned group of buildings. LANs are characterized by higher data-transfer rates, smaller geographic range, and lack of a need for leased telecommunication lines.



Local Area Network

Types of LANs:

1. **Wired LAN:**
 - o Uses cables and physical hardware (e.g., Ethernet cables).
 - o Commonly uses Ethernet standards (IEEE 802.3).
2. **Wireless LAN (WLAN):**
 - o Uses wireless signals (e.g., Wi-Fi).
 - o Commonly uses Wi-Fi standards (IEEE 802.11).
3. **Virtual LAN (VLAN):**

- o Uses software to group network devices into virtual segments, regardless of physical location.
- o Enhances network security and performance.

Network Equipment:

1. **Router:** Connects different networks and directs network traffic.



2. **Switch:** Connects devices within the same network and uses MAC addresses to forward data.



3. **Access Point (AP):** Allows wireless devices to connect to a wired network.



4. **Network Interface Card (NIC):** Hardware component that connects a computer to a network.



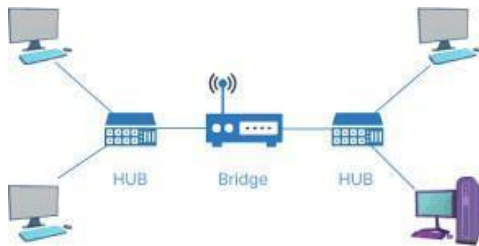
5. **Modem:** Converts digital data to analog signals and vice versa for data transmission over telephone lines.



6. **Repeater:** Amplifies or regenerates signals to extend the distance over which data can travel.



7. **Bridge:** Connects and filters traffic between two network segments.



8. **Firewall:** Monitors and controls incoming and outgoing network traffic based on security rules.



Steps of Installation / Configuration

1. Setting Up a Wired LAN:

- **Step 1:** Gather all necessary hardware (Ethernet cables, switches, routers, NICs).
- **Step 2:** Connect the router to the modem.
- **Step 3:** Connect the switch to the router.
- **Step 4:** Connect computers and other devices to the switch using Ethernet cables.
- **Step 5:** Configure the router settings via its web interface (usually accessed through a web browser).

- **Step 6:** Ensure all devices are configured with appropriate IP addresses, either dynamically (via DHCP) or statically.

2. Setting Up a Wireless LAN:

- **Step 1:** Gather all necessary hardware (Wi-Fi router, access points if needed).
- **Step 2:** Connect the Wi-Fi router to the modem.
- **Step 3:** Access the router's web interface to configure wireless settings (SSID, security settings).
- **Step 4:** Place access points in optimal locations to ensure coverage.
- **Step 5:** Connect devices to the Wi-Fi network using the configured SSID and password.

3. Configuring VLANs:

- **Step 1:** Access the switch's management interface (usually through a web browser).
- **Step 2:** Navigate to the VLAN configuration section.
- **Step 3:** Create new VLANs and assign ports to the respective VLANs.
- **Step 4:** Configure routers or layer 3 switches to route traffic between VLANs if needed.

Experiment- 02

Study and Verification of Standard Network Topologies

Lab Objective: Understanding and verifying the structure and functioning of various standard network topologies, including Star, Bus, and Ring.

Prerequisites: Basic knowledge of network topologies and networking devices.

Outcome: Gain practical experience in setting up and understanding different network topologies and their characteristics.

- **Step 1:** Gather all necessary hardware (computers, network interface cards, cables).
- **Step 2:** Connect each computer to the next one in the ring using the network cables, ensuring the last computer connects back to the first one.
- **Step 3:** Configure the network settings on each computer to ensure they are on the same network.

Experiment- 03

LAN Installation and Configuration

Lab Objective: Setting up and configuring a Local Area Network (LAN) to enable communication between multiple devices.

Prerequisites: Basic understanding of networking concepts and hardware.

Outcome: Successfully configuring a LAN, enabling network communication between connected devices.

Experiment- 04

Study of Transmission Media

Lab Objective: To study and understand different types of transmission media used in networking, including their characteristics and appropriate usage scenarios.

Prerequisites: Basic knowledge of networking concepts and understanding of data transmission principles.

Outcome: Identify and describe various types of transmission media, their characteristics, and appropriate usage scenarios.

Experiment- 05

Configuration and Use of Network Devices

Lab Objective: To configure and understand the functions of different network devices such as routers, switches, and hubs

Prerequisites: Basic understanding of networking concepts and familiarity with network configuration commands and tools.

Outcome: Successfully configure and demonstrate the use of routers, switches, and hubs in a network.

Experiment- 06

Setting Up a Wireless Network

Lab Objective: To set up and configure a wireless network using a wireless router and secure the network using appropriate security protocols.

Prerequisites: Basic understanding of networking concepts and familiarity with wireless network configuration.

Outcome: Successfully set up and configure a secure wireless network.

Description:

Wireless networks provide a convenient way for devices to communicate and access the internet without the need for physical connections. However, they also introduce security challenges that need to be addressed to protect the network from unauthorized access and data breaches.

Wireless Network Components:

Wireless Router: The central device that connects to the internet and allows wireless devices to communicate with each other and access the internet.

Devices: Computers, smartphones, tablets, and other wireless-enabled devices that connect to the network.

Access Points: Additional devices that extend the wireless coverage of the network.

Wireless Standards: Wireless networks operate based on different standards, which determine the speed and range of the network. The most common standards are:

802.11b: Operates at 2.4 GHz with a maximum speed of 11 Mbps.

802.11g: Operates at 2.4 GHz with a maximum speed of 54 Mbps.

802.11n: Operates at both 2.4 GHz and 5 GHz with a maximum speed of 600 Mbps.

802.11ac: Operates at 5 GHz with a maximum speed of up to several Gbps.

802.11ax (Wi-Fi 6): Operates at both 2.4 GHz and 5 GHz with improved efficiency and speed over previous standards.

Wireless Security Protocols: Securing a wireless network is crucial to prevent unauthorized access and protect data. The main wireless security protocols are:

WEP (Wired Equivalent Privacy):

- o **Description:** The original wireless security standard.
- o **Characteristics:** Uses static encryption keys. Vulnerable to various types of attacks due to weak encryption.
- o **Usage:** Considered obsolete and not recommended for use.

WPA (Wi-Fi Protected Access):

- o **Description:** A security protocol developed to address the weaknesses of WEP.
- o **Characteristics:** Uses TKIP (Temporal Key Integrity Protocol) for dynamic key generation.
- o **Usage:** Provides better security than WEP but has been largely replaced by WPA2.

WPA2 (Wi-Fi Protected Access 2):

- o **Description:** The current widely used wireless security standard.
- o **Characteristics:** Uses AES (Advanced Encryption Standard) for strong encryption. Supports both personal (pre-shared key) and enterprise (RADIUS server) modes.
- o **Usage:** Recommended for most networks due to its strong security features.

WPA3 (Wi-Fi Protected Access 3):

- o **Description:** The latest security protocol designed to provide enhanced security over WPA2.
- o **Characteristics:** Provides better protection against brute-force attacks, improved encryption for open networks, and individualized data encryption.
- o **Usage:** Recommended for new networks were supported by devices.

Experiment- 07

Setting Up a Wireless Network

Lab Objective: To configure and verify IPv4 and IPv6 addressing on network devices.

Prerequisites: Basic understanding of IP addressing and familiarity with network configuration commands.

Outcome: Successfully configure and verify IPv4 and IPv6 addressing.

Experiment- 08

Subnetting and Creating Subnet Masks

Lab Objective: To perform subnetting on a given IP address and create subnet masks.

Prerequisites: Basic understanding of IP addressing and subnetting concepts.

Outcome: Successfully configure and verify IPv4 and IPv6 addressing.

Experiment- 09

Configuring Remote Access Connections

Lab Objective: To set up and configure remote access connections using different technologies.

Prerequisites: Basic understanding of networking and remote access technologies and familiarity with configuring network devices and remote access protocols.

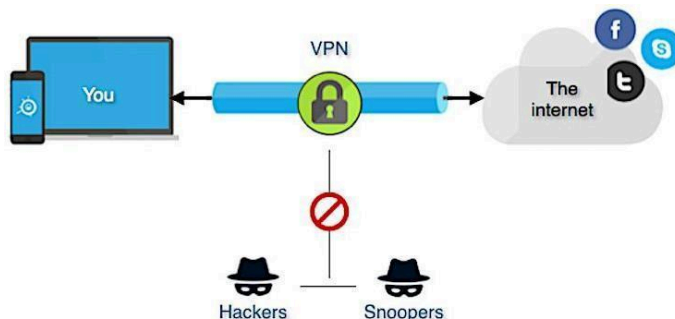
Outcome: Successfully create subnets and configure subnet masks.

Description:

Remote access allows users to connect to a network or computer from a distant location. This is particularly useful for accessing resources, performing maintenance, or providing support from remote locations. Various technologies facilitate remote access, including VPN (Virtual Private Network), RDP (Remote Desktop Protocol), SSH (Secure Shell), and more.

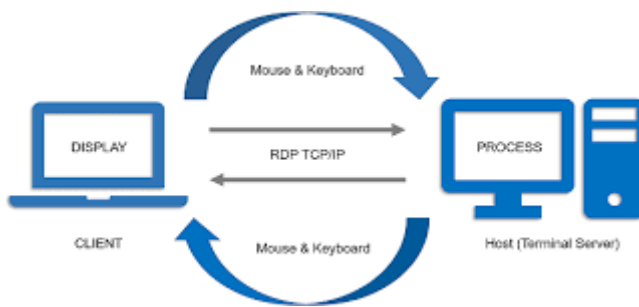
Key Remote Access Technologies:

1. VPN (Virtual Private Network):

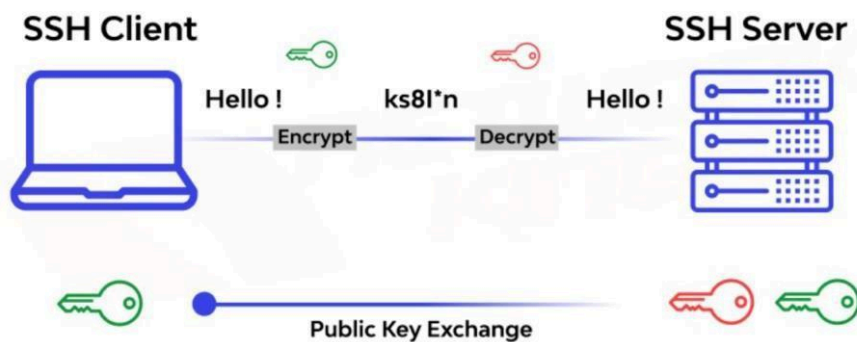


- **Description:** VPNs create a secure tunnel over the internet to connect remote users to a private network. This ensures data privacy and security.
- **Types:**
 - **Site-to-Site VPN:** Connects entire networks to each other.
 - **Remote Access VPN:** Allows individual users to connect to a private network.

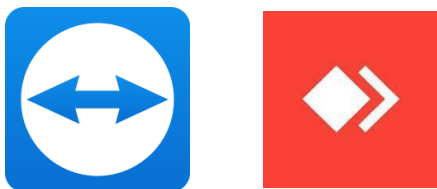
2. RDP (Remote Desktop Protocol):



- o **Description:** RDP, developed by Microsoft, allows users to connect to and control a computer remotely. It is commonly used for accessing Windows machines.
 - o **Features:** Graphical interface, file transfer, and remote application access.
- ## 3. SSH (Secure Shell):

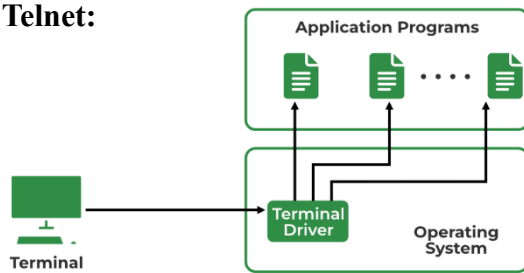


- o **Description:** SSH is a cryptographic network protocol used for secure data communication, remote shell services, and command execution.
 - o **Features:** Secure connection, port forwarding, and secure file transfer (SFTP).
- ## 4. TeamViewer and AnyDesk:



- o **Description:** These are third-party remote desktop software solutions that facilitate remote access to computers, file sharing, and collaboration.
- o **Features:** Cross-platform support, easy setup, and strong encryption.

5. Telnet:



- o **Description:** Telnet is an older protocol for remote access but lacks security features, as data is transmitted in plain text. It is mostly replaced by SSH.
- o **Use Case:** Primarily used for legacy systems.

Security Considerations:

- Use strong encryption protocols (e.g., SSL/TLS for VPN, SSH).
- Implement multi-factor authentication (MFA).
- Regularly update and patch remote access software.
- Use strong, unique passwords for remote access accounts.
- Restrict remote access to authorized users and IP addresses.

Experiment- 10

Configuring Remote Access Connections

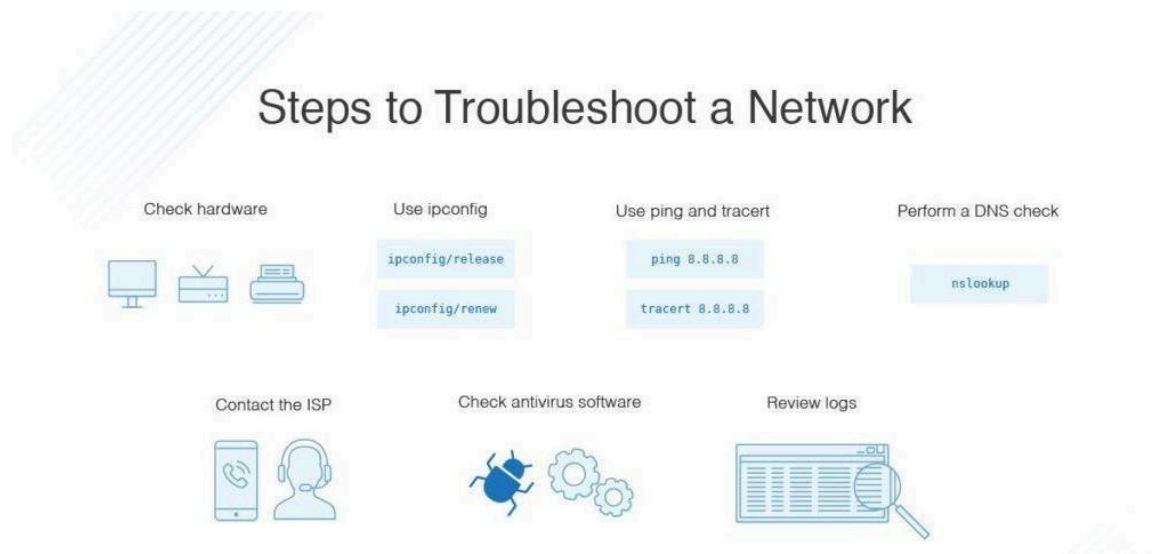
Lab Objective: To use various network troubleshooting tools to diagnose and resolve network issues.

Prerequisites: Basic understanding of networking concepts and familiarity with command-line interface tools.

Outcome: Successfully use network troubleshooting tools to identify and resolve network problems.

Description:

Network troubleshooting involves identifying, diagnosing, and resolving issues within a network. Various tools are available to assist in this process, each designed to target specific types of problems. Understanding how to use these tools effectively is crucial for maintaining a stable and efficient network.



Common Network Troubleshooting Tools:

1. Ping:

- o **Description:** Ping is a command-line utility used to test the reachability of a host on an IP network. It sends ICMP Echo Request messages and waits for Echo Reply messages.
- o **Use Case:** Checking connectivity between devices and determining round-trip time.

2. Traceroute/Tracert:

- o **Description:** Traceroute (or tracert in Windows) is a network diagnostic tool used to track the path that packets take to reach a destination. It displays each hop along the route.
- o **Use Case:** Identifying the path and delays of packets traveling through a network.

3. IPconfig/Ifconfig:

- o **Description:** IPconfig (Windows) and Ifconfig (Linux) are command-line utilities used to display and configure the network interface settings of a device.
- o **Use Case:** Viewing current network configuration, renewing DHCP leases, and troubleshooting IP issues.

4. Netstat:

- o **Description:** Netstat is a command-line tool that displays network connections, routing tables, interface statistics, masquerade connections, and multicast memberships.
- o **Use Case:** Monitoring active network connections and diagnosing network performance issues.

5. Nslookup:

- o **Description:** Nslookup is a network administration command-line tool used for querying the Domain Name System (DNS) to obtain domain name or IP address mapping.
- o **Use Case:** Troubleshooting DNS resolution issues.

6. Wireshark:

- o **Description:** Wireshark is a network protocol analyzer that captures and displays data traveling back and forth on a network in real-time.
- o **Use Case:** Deep packet inspection to analyze network traffic and diagnose complex issues.

7. Pathping:

- o **Description:** Pathping is a network utility in Windows that combines the functionality of ping and tracert. It provides information about network latency and packet loss at each hop.
- o **Use Case:** Diagnosing packet loss and latency issues in a network path.

8. Telnet/SSH:

- o **Description:** Telnet and SSH are protocols used to access and manage devices remotely over a network. SSH is secure, while Telnet is not.
- o **Use Case:** Remote management and troubleshooting of network devices.